

HW 3 - Grading - rubrics (by JK, 2019-10-30)

Part 1: Population growth in a 2-compartment model

- **Question 1 (1 point):** explaining change of resources, like: "The rate of change of resources is proportional to ..."
- **Question 2 (3 points):** 1 point for each answer:
 - 1 pt for having 2 ODEs
 - 1 pt for correctly writing each ODE
- **Question 3 (4 points):**
 - 2 pts for correctly implementing ODEs
 - 1 pt for accepting right parameters, 1 pt for returning 2 values
- **Question 4 (4 point):**
 - 1 pt for correct time
 - 1 pt for correct call to `odeint`
 - 1 pt for each plot
- **Question 5 (3 point):**
 - 2 pt for plot, including labels
 - 1 pt for observation

Part 2: Population growth in a 3-compartment model

- **Question 6 (2 point):** answer indicating current model is basically simpler model from *part 1*.
 - **Question 7 (5 point):**
 - 3 pt for correctly implemented ODEs
 - 1 pts accepting 3 values
 - 1 pts returning 3 values
 - **Question 8 (4 points):**
 - 1pt for correctly running the `odeint`
 - 2 pts for correct plots
 - 1 pt for observation, the answer should indicate that: "the results are the same as in Part 1 because $b=0$ "
 - **Question 9 (4 point):**
 - 1pt for correctly running the `odeint`
 - 2 pts for correct plots
 - 1 pt for observation, the answer should indicate that: "population at large time is proportional to 'b' (linearly growing for $b>0$)."
 - **Question 10 (5 point):**
 - 2 pts for computing both scenarios
 - 1 pt for plot of both scenarios
 - 2 pt for answering both questions: Q1: 1st scenario is better, Q2: both are OK
 - **Question 11 (5 point):**
 - 2 pts for computing both scenarios
 - 1 pt for plot of both scenarios
 - 2 pt for answering both questions: Q1: 1st scenario is better, Q2: 2nd scenario is better
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